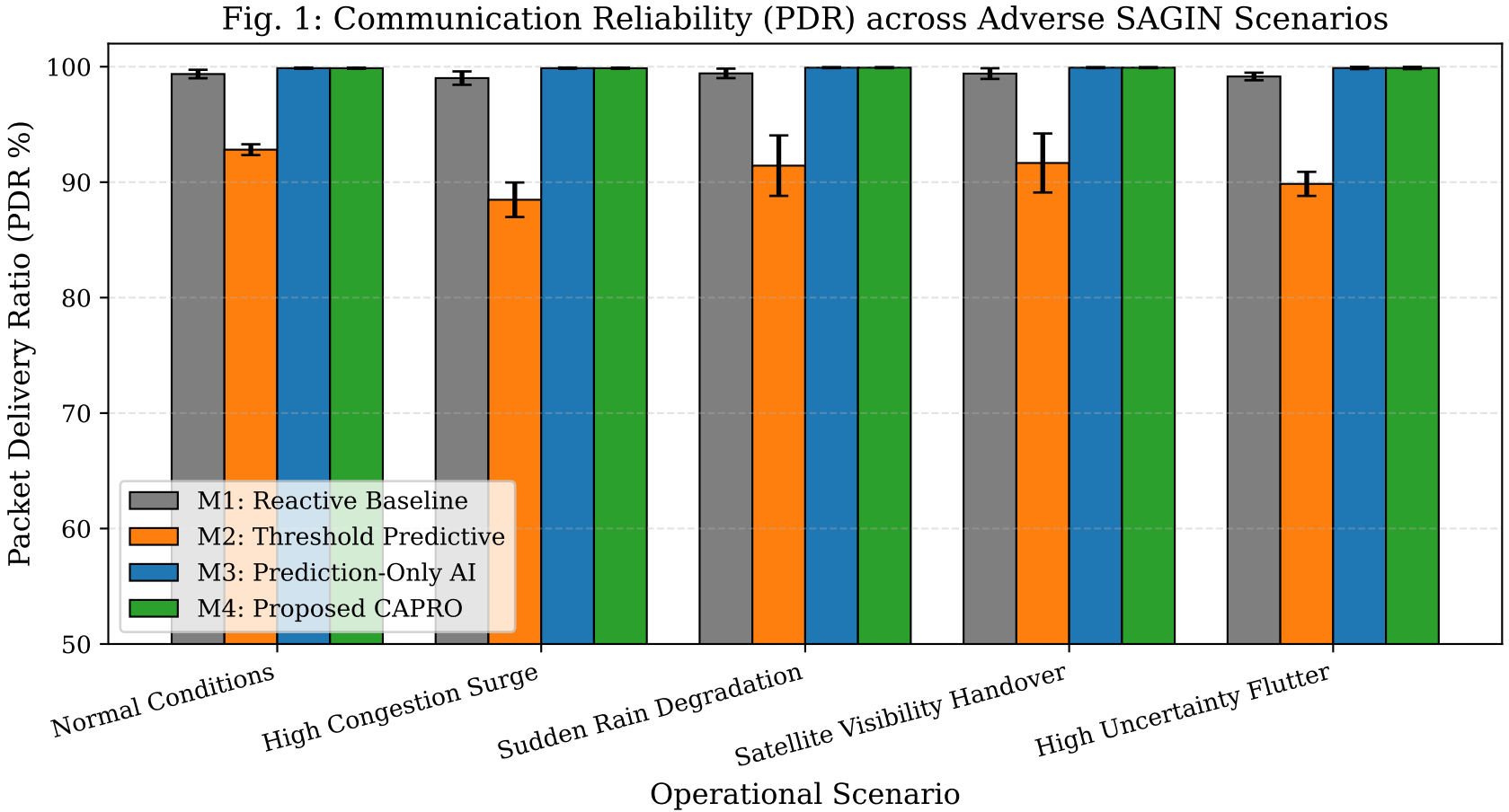
Fig. 1: Communication Reliability (PDR) across Adverse SAGIN Scenarios
Packet Delivery Ratio (PDR %)
100
90
80
70
60
50
M1: Reactive Baseline
M2: Threshold Predictive
M3: Prediction-Only AI
M4: Proposed CAPRO
Normal Conditions
High Congestion Surge
Sudden Rain Degradation
Satellite Visibility Handover
High Uncertainty Flutter
Operational Scenario